# Mohamed FATTA

**AI Engineer / Full Stack Developer — Agentic AI & GenAI | Java/Spring, Angular · AWS**

Agadir, Morocco | medfata3@gmail.com | +212 638653350
Portfolio | LinkedIn | GitHub

## PROFILE

Full Stack Developer with 4+ years of experience building cloud-native AWS applications (serverless and event-driven architectures, Java/Spring, Angular), now focused on designing AI agents and agentic workflows: RAG pipelines, multi-LLM orchestration and routing, and exposing agents through the MCP protocol. As a freelancer, I deliver end-to-end AI automations for clients, from scoping to production, and enjoy helping non-technical teams adopt these tools. Core stack: Python, Java/Spring, TypeScript/Node.js, Angular, AWS.

## PROFESSIONAL EXPERIENCE

### Freelance Full Stack Developer *2025 – Present · Remote*

- Delivered end-to-end custom solutions for clients, from scoping to production, including AI-agentic automations (agents, RAG pipelines, LLM API integration).
- Built a custom bundle module for postcards.shop to drive upsell and average order value (AOV).
- Developed an AI-based automation and a site-wide theme redesign for honeyswim.co.
- Designed and shipped Bundlora, a Shopify product-bundling app, landing its first paying customer.

*Python/FastAPI, TypeScript, Node.js, React/Next.js, AI agents (LLM), MCP, Shopify (Liquid, Apps), Supabase.*

### Software Engineer — NTT DATA *February 2023 – May 2025, Tetouan*

***SKY OSS Project***

- Designed a serverless AWS solution for real-time notifications via WebSockets, serving several thousand users.
- Developed and debugged Lambda functions; wrote unit tests (Pytest, moto).
- Improved Lambda performance by 70% by reducing Layers and optimizing DB connections.
- Event-driven architecture (API Gateway, DynamoDB, SQS, SNS); IaC via AWS CDK.

*Python, Pytest/moto, boto3, AWS Lambda Powertools, AWS (IAM, Lambda, API Gateway, DynamoDB, SQS, SNS, CloudWatch), AWS CDK.*

***Ferrovial Project — highway tolling and tracking system (Portugal)***

- Debugged a production tolling system processing tens of thousands of transactions per day (Java/Spring).
- Fixed critical bugs in payment services (Kafka, dead-letter queues); resolved a double-billing issue via idempotent deduplication.
- Application-level debugging and deployment triggers in a Kubernetes environment; occasional contributions to the Angular front end.

*Java, Spring, Spring Batch, MongoDB, Kafka, Kubernetes, Angular.*

***GO-DISTRIBUTION Project***

- Batch processing (Spring Batch) syncing several thousand records per day from a third-party system; built reusable Angular components.

*Angular, TypeScript, Java, Spring Boot, PostgreSQL.*

### Software Engineer — FY Computing *October 2021 – January 2023, Rabat*

- Developed a SaaS platform automating application deployment for other organizations, with multi-tenant architecture and per-organization data isolation.
- Fine-grained IAM solution (customized Keycloak); secured application secrets via HashiCorp Vault.
- Generic multi-cloud file management REST API (S3, local disk); end-to-end tests (Robot Framework).

*Angular, TypeScript, Keycloak, HashiCorp Vault, Robot Framework, Python, AWS (S3, EC2).*

### Internship — Software Developer — WEPILOT *September 2020 – September 2021, Agadir*

- Full-stack development of a budget management application for a university in Agadir.

*Angular, TypeScript, Java, Spring Boot, MySQL.*

## PROJECTS — AGENTIC AI & GENAI

### RTFM For Me Agent — AI documentation agent (RAG) *2026*

- Full RAG pipeline (ingestion, embeddings, semantic cache, citations) with an intent router that distinguishes documentation questions, chitchat, memory recall, and actions.
- Exposed the agent as an MCP server (5 tools) so it can be driven by external agent clients (Claude Code/Desktop, Cursor, etc.).

- Multi-model strategy (fast / generation / economy lanes) with automatic failover on errors or exceeded quotas.
- Multi-tenancy, document versioning with staleness detection, and web crawling with human review before content is added to the corpus.

*Python, FastAPI, Redis 8 (vector search), multi-provider LLMs (Groq, Gemini), MCP protocol — github.com/medfata/codin-challenge-rtfm-agent*

**BYO MCP Server — MCP server for AI agents** *2026*

- MCP server exposing tools that let AI agents find code solutions in a shared GitHub repository.
- Self-documentation tool that generates a codebase overview from the project's own signatures and comments.

*TypeScript, Node.js, MCP protocol (official SDK) — github.com/medfata/byo-mcp-server*

**SourceMax AI — LLM-based YouTube channel analysis** *2025 – Present*

- Map-reduce pipeline for summarizing long-form content via LLM.

*Python/FastAPI, React.*

## TECHNICAL SKILLS

**Generative AI & Agents:** AI agents & agentic workflows, RAG, Model Context Protocol (MCP), multi-model orchestration and failover, prompting & structured outputs, LLM API integration (OpenAI-compatible, Groq, Gemini), Redis (vector search).

**Backend & Cloud:** Python, FastAPI, Java, Spring Boot, Node.js, serverless & event-driven architecture, REST APIs, AWS (Lambda, API Gateway, DynamoDB, SQS, SNS, S3, EC2, IAM, CloudWatch), AWS CDK.

**Frontend:** Angular, React/Next.js, TypeScript, JavaScript, HTML/CSS.

**Databases:** PostgreSQL, MySQL, DynamoDB, MongoDB, Redis.

**Security & Authentication:** Keycloak, HashiCorp Vault, IAM.

**Testing & Quality:** Pytest, moto, Robot Framework, unit & e2e testing, code reviews.

**Tools & DevOps:** Git, Linux, Docker, Kubernetes, Jira/Confluence.

## SOFT SKILLS

Problem solving | Continuous learning | Technical communication | Explaining concepts & supporting teams | Autonomy

## EDUCATION

**OFPPT — ISTA** *2017 – 2019, Agadir*
Specialized Technician in IT Development

## LANGUAGES

Arabic: native | French: professional (B2) | English: professional (B2)